Open Virtual Research Environment (OpenVRE)

An interoperable Virtual Research Environment for secure data access, scalable computation, and collaboration

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Abstract

The rapid growth of genomic and biological data, combined with the complexity of biomedical workflows, has created challenges in achieving interoperability across High-Performance Computing (HPC), Cloud Computing facilities, and secure data platforms. Seamless integration, robust security, and efficient resource utilization are crucial for enabling the optimal use of health data.

In this poster, we present OpenVRE (Open Virtual Research Environment), an open-source, cloud-based platform designed to address these challenges. OpenVRE bridges the gap between HPC resources, sensitive data infrastructures, and analytical tools and workflows, providing a flexible environment for researchers.

Key features

Customizable design

Modular and flexible interface that adapts to different research workflows and user needs.

Data management and integration

Supports diverse data types with built-in storage, metadata handling, and seamless dataset organization.

Integrated analysis tools

Easily connect and run analytical tools locally, on HPC systems, or through cloud services.

Interoperability

Connects with external databases and repositories to enrich and streamline research workflows.

Integration into Trusted Research Environment (TRE)

Facilitates secure data access and analysis while ensuring compliance with regulatory frameworks.

Architecture

Core components

Virtual Research Environment

Digital workspace designed to support the entire lifecycle of a research project. It provides researchers with the tools, services, and resources needed to conduct collaborative research online.

Database

Stores all data of the VRE, including user information, job metadata and tool configurations. Its flexible schema allows for dynamic data modeling, making it ideal for the varying requirements of computational tasks.

Authentication and Authorization Server (Login)

Keycloak serves as the authorization and authentication component for openVRE. It enables secure user authentication and role-based access control.

Job scheduler

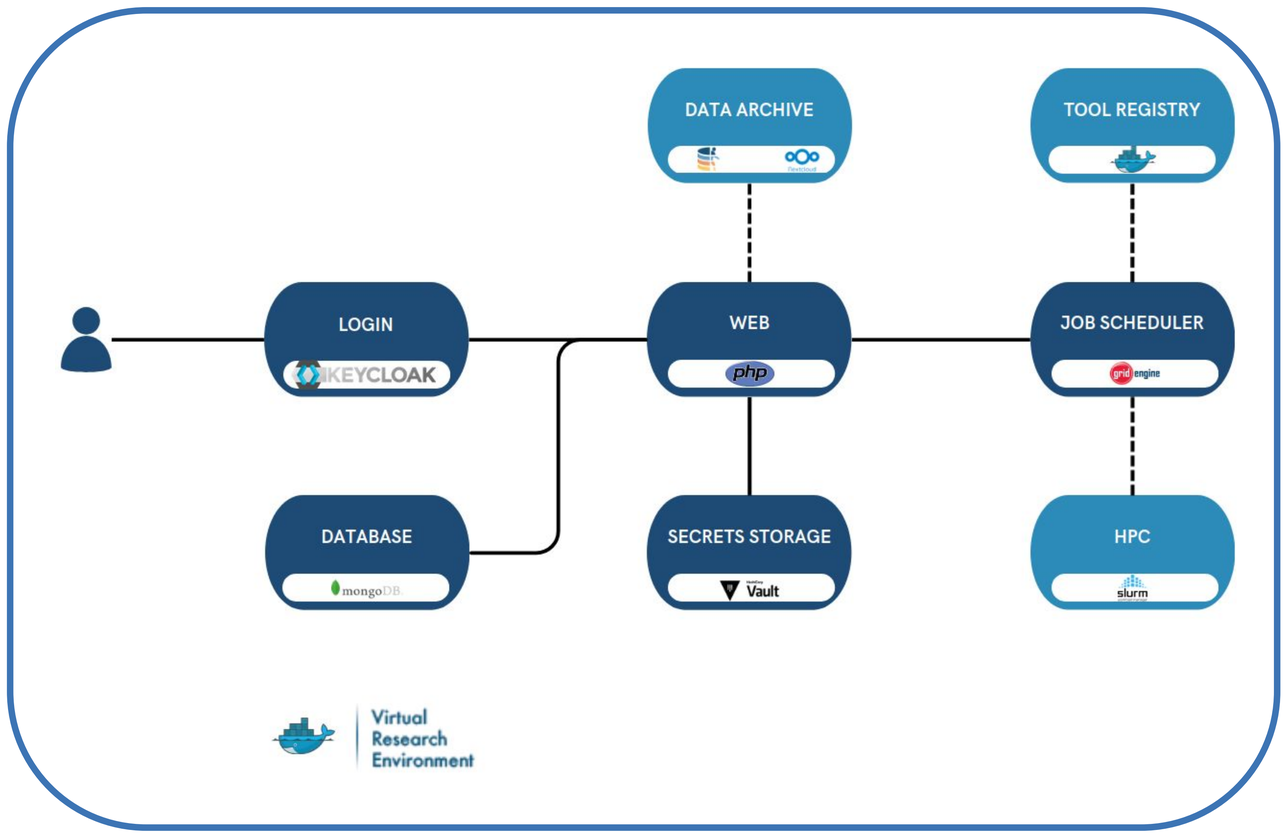
Handles job scheduling and resource allocation. Currently support is given for Grid Engine.

Web service

Interface to facilitate the user with the tasks: finding data, organizing it, executing workflows, downloading results, etc.

Secrets storage

HashiCorp Vault is used to securely store keys and credentials that users need to launch jobs or access computational environments and databases.



External components

Data archive

External databases and repositories which are connected to the VRE (e.g. EGA and Nextcloud). Data can be also locally uploaded.

Tool registry

Set of dockerized analysis tools and workflows.

High-Performance Computing

Allows the execution of computationally expensive jobs on remote premises (e.g. MareNostrum 5).

Resources

Main repository

Wiki

Instances in European projects

EOSC4Cancer VRE

GDI VRE

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